

Multiple Choice (10 marks)

1. What is penetration testing?
 - (a) A procedure for testing libraries or other program components for vulnerabilities
 - (b) Whole-system testing for security flaws and bugs
 - (c) A security-minded form of unit testing that applies early in the development process
 - (d) All of the above
2. Message authentication is a service beyond
 - (a) Message Confidentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
3. The access matrix approach to protection has the difficulty that
 - (a) the matrix, if stored directly, is large and can be clumsy to manage
 - (b) it is not capable of expressing complex protection requirements
 - (c) deciding whether a process has access to a resource is undecidable
 - (d) there is no way to express who has rights to change the access matrix itself
4. A _____ may be a hidden part of a program, a separate infected program a Trojan in disguise of an executable or code in the firmware of any system's hardware.
 - (a) crypter
 - (b) virus
 - (c) backdoor
 - (d) key-logger
5. How is IP address spoofing detected?
 - (a) Installing and configuring a IDS that can read the IP header
 - (b) Comparing the TTL values of the actual and spoofed addresses
 - (c) Implementing a firewall to the network
 - (d) Identify all TCP sessions that are initiated but does not complete successfully

True / False (10 marks)

Answer True or False

6. True or False: Exploitation of the Heartbleed bug allows an attacker to read memory outside the bounds of a buffer.
7. True or False: Brute-force attacks on access credentials primarily occur at the session layer of the OSI model.
8. True or False: The message must be decrypted at the sender site after being encrypted at the receiver site.
9. True or False: Trojans that exploit vulnerabilities in software are primarily designed to steal personal information without incurring costs.
10. True or False: The TCP connect scan does not fully establish a TCP connection, similar to the SYN stealth scan.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to return true if the given number is even else return false.
12. Write a function to count the elements in a list until an element is a tuple.

End of Paper